SIMATS ENGINEERING

Lab Programs

COURSE CODE: ECA1009 **COURSE NAME: Microprocessors & Microcontrollers**

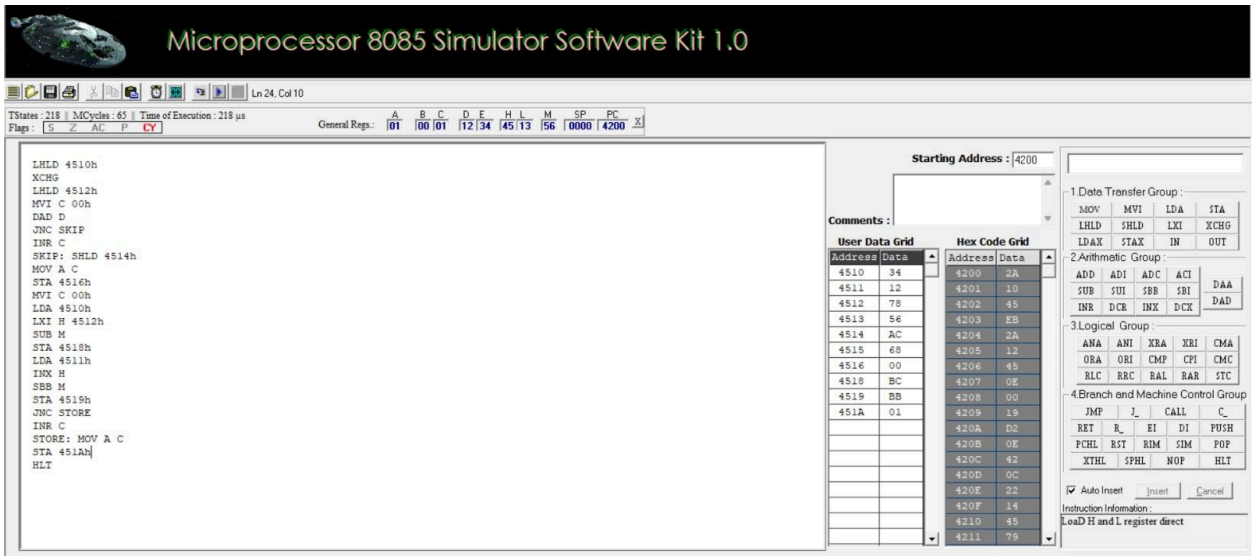
***Program B* – Design and Verification of 16-Bit Addition and Subtraction Using Intel 8-Bit Microprocessor Assembly Language Programming**

Aim: To design and execute an 8085 assembly language program for 16-bit addition and subtraction using an 8-bit microprocessor.

Algorithm:

1. Load the 16-bit operands from the specified memory locations into register pairs.
2. Exchange the operands using the XCHG instruction wherever required.
3. Perform 16-bit addition using DAD and handle the carry using conditional branching.
4. Store the addition result in the specified memory locations.
5. Perform 16-bit subtraction using SUB and SBB to account for borrow.
6. Store the subtraction result and halt the program.

Execute:



Result: The 16-bit addition and subtraction program was executed successfully using the 8085 simulator, and the results were stored in the specified memory locations.